even at ordinary, everyday speeds, will have increasingly different conceptions of *now* if they are increasingly far apart in space.

To make this concrete, imagine that Chewie is on a planet in a galaxy far, far away—10 billion light-years from earth—idly sitting in his living room. Imagine further that you (sitting still, reading these words) and Chewie are not moving relative to each other (for simplicity, ignore the motion of the planets, the expansion of the universe, gravitational effects, and so on). Since you are at rest relative to each other, you and Chewie agree fully on issues of space and time: you would slice up spacetime in an identical manner, and so your now-lists would coincide exactly. After a little while, Chewie stands up and goes for a walk—a gentle, relaxing amble—in a direction that turns out to be directly away from you.

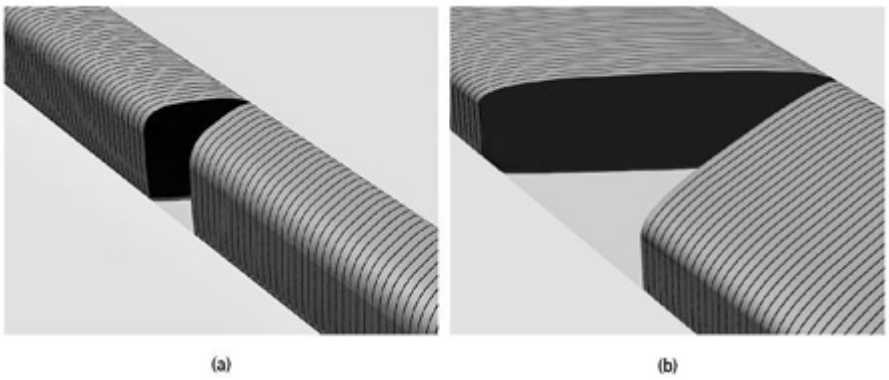


Figure 5.2 (a) In an ordinary loaf, slices cut at slightly different angles don’t separate significantly. (b) But the larger the loaf, for the same angle, the greater the separation.

This change in Chewie’s state of motion means that his conception of *now,* his slicing up of spacetime, will rotate slightly (see Figure 5.3). This tiny angular change has no noticeable effect in Chewie’s vicinity: the difference between his new *now* and that of anyone still sitting in his living room is minuscule. But over the enormous distance of 10 billion light years, this tiny shift in Chewie’s notion of *now* is amplified (as in the passage from Figure 5.3a to 5.3b, but with the protagonists now being a huge distance apart, significantly accentuating the shift in their *nows*). *His now and your now, which were one and the same while he was sitting still, jump apart because of his modest motion.*